

CSCA 考纲拆解

- 数学篇

Mathematics Examination Syllabus Analysis

题目数量 Total Questions

48 MCQs

考试时长 Duration

60 Minutes

总分 Total Score

100



Good luck guys~



Beijing Crossline Edu Tech Co., Ltd.

Exam Overview

Overview of Exam Structure and Requirements

Purpose:	Assess students' mastery of core mathematical concepts and problem-solving abilities
Question Type:	Multiple Choice (4 options)
Duration:	60 minutes
Language:	Bilingual (Chinese/English)
Total Questions:	48 Questions
Total Points:	100 Points
Average Time per Question:	1.25 minutes (75 seconds)



Exam Features

The CSCA Mathematics Exam 2025 is designed to **comprehensively assess** students' mathematical abilities, covering four major modules with balanced distribution.

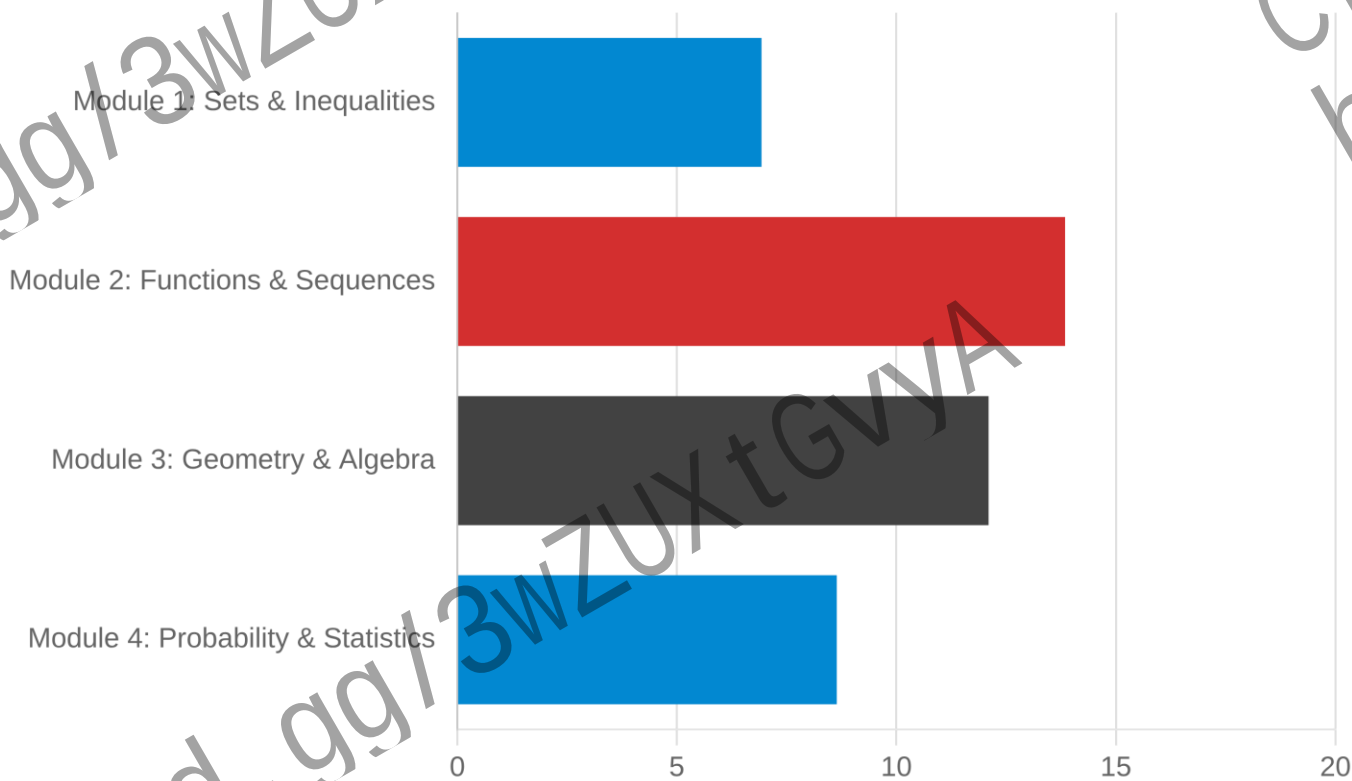
The exam emphasizes **practical application** and **logical reasoning**, requiring students to not only master basic concepts but also apply them flexibly to solve problems.

The difficulty distribution follows a **40%-30%-30% structure** (Easy-Medium-Complex), ensuring comprehensive evaluation of students at different levels.

All questions must be **original**, with clear labeling of module, difficulty, answer, and explanation.

Four Core Modules Distribution

Distribution of Four Core Modules - Question Count and Percentage



Module Distribution Features

Module 2: Functions & Sequences has the highest proportion at 33%, reflecting its importance in the mathematics curriculum.

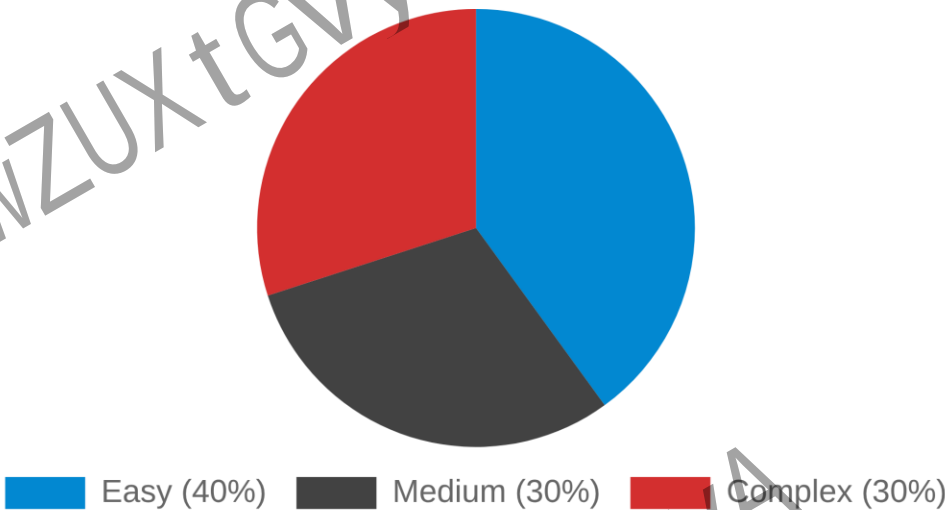
Module 3: Geometry & Algebra accounts for 29%, emphasizing spatial thinking and algebraic reasoning.

Module 4: Probability & Statistics accounts for 21%, focusing on data analysis and statistical applications.

Module 1: Sets & Inequalities accounts for 17%, covering foundational concepts.

Difficulty Distribution Analysis

Analysis of Difficulty Distribution Across Modules



Difficulty Structure by Module

Module	Easy	Medium	Complex
Sets & Inequalities	4 questions	2 questions	2 questions
Functions & Sequences	6 questions	5 questions	5 questions
Geometry & Algebra	5 questions	5 questions	4 questions
Probability & Statistics	5 questions	3 questions	2 questions



Difficulty Design Philosophy

Easy Questions (40%) - Focus on basic concepts and direct application of formulas, ensuring students can demonstrate foundational knowledge.

Medium Questions (30%) - Require integrated reasoning and multi-step problem solving, testing students' ability to connect different concepts.

Complex Questions (30%) - Involve comprehensive analysis and advanced problem-solving strategies, distinguishing high-achieving students.

This **40-30-30 distribution** ensures comprehensive assessment while maintaining appropriate challenge levels.

Each module maintains similar difficulty proportions to ensure balanced evaluation across all content areas.

Module 1: Sets and Inequalities

Module 1: Sets and Inequalities | 8 Questions (17%)

Knowledge Points

Set Operations: Union, intersection, complement, and their properties

Linear Inequalities: Solving and graphing linear inequalities in one or two variables

Absolute Value Inequalities: Solving inequalities involving absolute values

Question Requirements

• Easy (4 questions)

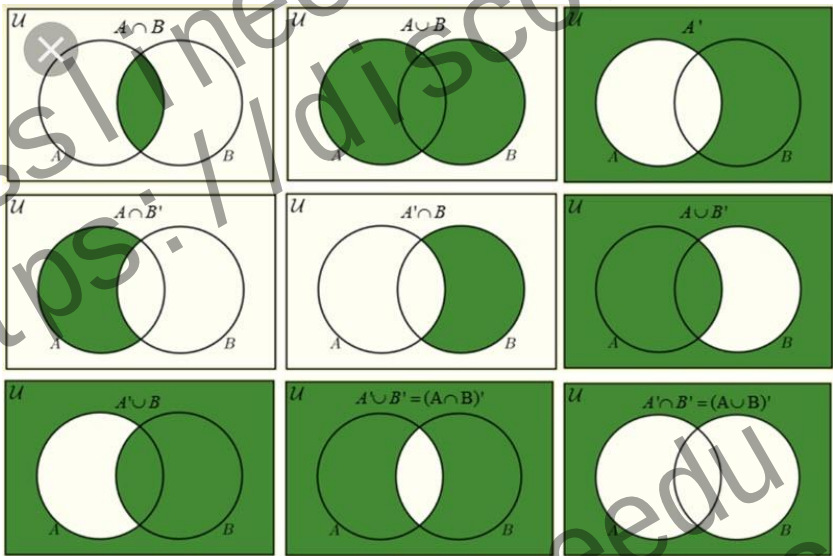
Basic set operations, such as finding unions or intersections; solving simple linear inequalities

• Medium (2 questions)

Combined set operations, such as problems involving multiple sets; solving compound inequalities

• Complex (2 questions)

Set problems involving parameters or abstract reasoning; solving complex absolute value inequalities or systems of inequalities



Module Features

Sets and Inequalities module accounts for 17%, serving as the **foundational module** of the entire exam.

This module emphasizes **logical reasoning** and **abstract thinking**, requiring students to master set notation and inequality solving techniques.

Key focus areas include **set operations**, **linear inequalities**, and **absolute value inequalities**.

Difficulty structure: **4 Easy / 2 Medium / 2 Complex**

Module 2: Functions and Sequences

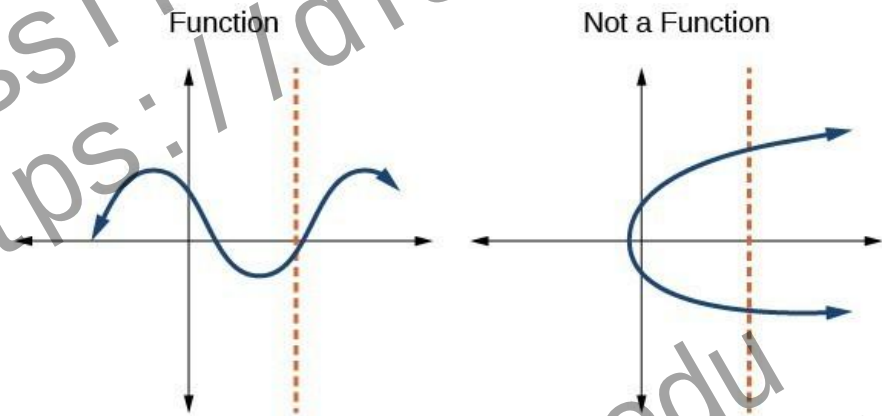
Module 2: Functions and Sequences | 16 Questions (33%)

Knowledge Points

- Function Properties:** Domain, range, monotonicity, parity, periodicity
- Specific Functions:** Linear, quadratic, exponential, logarithmic functions and their properties
- Sequences:** Arithmetic and geometric sequences, general term formulas, sum formulas
- Derivatives:** Basic derivative rules, applications in finding extrema and monotonicity

Question Requirements

- Easy (6 questions)**
Basic function properties, such as finding domain or range; simple sequence calculations
- Medium (5 questions)**
Function composition and transformations; solving sequence problems involving multiple steps
- Complex (5 questions)**
Complex function analysis involving parameters; advanced sequence problems with recursive formulas or summations; derivative applications



Module Features

Functions and Sequences module accounts for 33%, making it the **largest and most important module** in the exam.

This module emphasizes **analytical thinking** and **computational skills**, requiring students to master various function types and sequence patterns.

Key focus areas include **function properties**, **exponential and logarithmic functions**, and **arithmetic/geometric sequences**.

Difficulty structure: **6 Easy / 5 Medium / 5 Complex**

Module 3: Geometry and Algebra

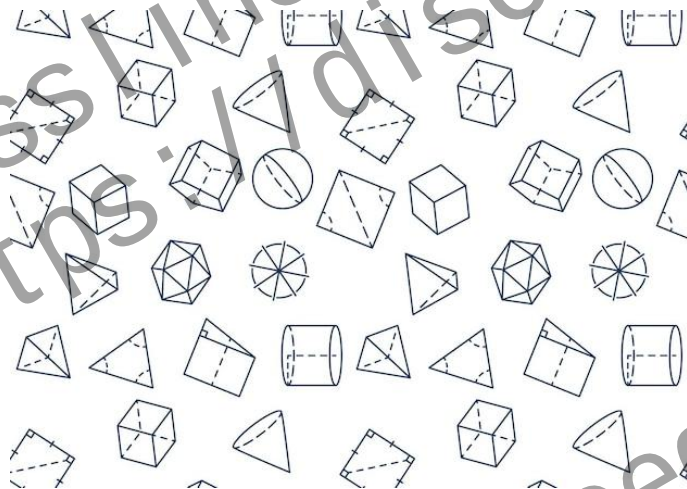
Module 3: Geometry and Algebra | 14 Questions (29%)

Knowledge Points

- Plane Geometry:** Properties of triangles, circles, and polygons; area and perimeter calculations
- Analytic Geometry:** Lines, circles, parabolas, ellipses, and hyperbolas in the coordinate plane
- Vectors:** Vector operations, dot product, applications in geometry
- Linear Algebra:** Systems of linear equations, matrices, determinants

Question Requirements

- Easy (5 questions)**
Basic geometric properties and calculations; simple linear equations and vector operations
- Medium (5 questions)**
Integrated geometry-algebra problems; analytic geometry involving conic sections; systems of equations
- Complex (4 questions)**
Complex geometric proofs or constructions; advanced conic section problems; matrix applications and transformations



Module Features

Geometry and Algebra module accounts for 29%, making it the **second largest module** in the exam.

This module emphasizes **spatial reasoning** and **algebraic manipulation**, requiring students to integrate geometric and algebraic thinking.

Key focus areas include **analytic geometry**, **conic sections**, and **vector applications**.

Difficulty structure: **5 Easy / 5 Medium / 4 Complex**

Module 4: Probability and Statistics

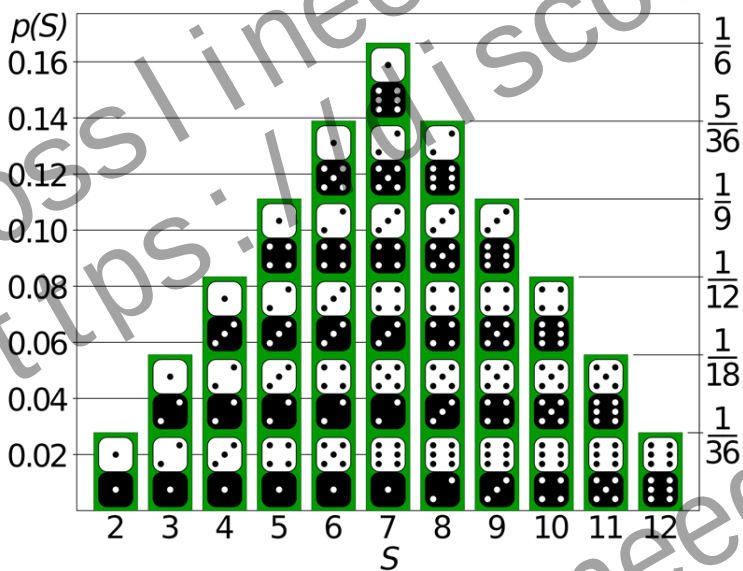
Module 4: Probability and Statistics | 10 Questions (21%)

Knowledge Points

- Classical Probability:** Basic events, sample space, probability calculation formulas
- Mean and Variance:** Measures of central tendency and dispersion of data
- Normal Distribution:** Probability characteristics of normal distribution, standard normal distribution, interval probability calculations

Question Requirements

- Easy (5 questions)**
Basic probability calculations, such as simple event probabilities; basic statistical measures (mean, variance) calculations
- Medium (3 questions)**
Multi-event combinations, such as conditional probability, independent event probability calculations; comprehensive statistical data analysis
- Complex (2 questions)**
Problems involving data analysis or multi-step reasoning, such as standard deviation comparisons, normal interval probability calculations, complex probability model construction and solving



Module Features

Probability and Statistics module accounts for 21%, the smallest of the four modules, but has the **strongest practical application**.

This module emphasizes **data analysis skills** and **statistical thinking**, requiring students to extract information from data and make reasonable inferences.

Key focus areas include **classical probability** basic calculations and **normal distribution** properties and applications.

Difficulty structure: **5 Easy / 3 Medium / 2 Complex**